13 Lecture 13

13.1 Moment generating functions

Definition 13.1. Suppose that X is random variable. The *moment generating function* of X, $M_X \colon \mathbb{R} \to (0, +\infty]$ is defined as

$$M_X(t) = \mathbb{E}[e^{tX}].$$

By taking into account Taylor's expansion $e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!}$, we have

$$M_X(t) = \mathbb{E}\left[\sum_{n=0}^{+\infty} \frac{t^n X^n}{n!}\right] = \sum_{n=0}^{+\infty} \frac{t^n}{n!}\mathbb{E}[X^n].$$

The numbers $\mathbb{E}[X^n],\ n \in \mathbb{N}$ are called *moments* of the random variable X. The following proposition justifies the name "moment generating function".

Proposition 13.2. *Assume that X, Y are two random variables with the same distribution $X \sim Y$ and $\mathbb{E}[|X|^n] < \infty$, for some $n \in \mathbb{N}$. Then,*

1. *$M_X(t) = M_Y(t)$ for all $t \in \mathbb{R}$.*
2. *$\mathbb{E}[X^n] = \frac{d^n M_X}{dt^n}(0)$.*

Proof. 1. $\mathbb{E}[e^{tX}] = \int e^{tX} d\mathbb{P} = \int e^{tx} d\mathbb{P}^X(x) = \int e^{tx} d\mathbb{P}^Y(x) = \mathbb{E}[e^{tY}]$.

2.
$$\frac{d^n M_X}{dt^n}(t) = \mathbb{E}[X^n e^{tX}] \overset{t=0}{\Longrightarrow} \mathbb{E}[X^n] = \frac{d^n M_X}{dt^n}(0).$$

□

We compute the moment generating functions of the distributions we studied in Lectures 7 and 8

Proposition 13.3. *Let X be a random variable.*

1. *If $X \sim Ber(p)$, then $M_X(t) = 1 - p + pe^t$.*
2. *If $X \sim Bin(n, p)$, then $M_X(t) = (1 - p + pe^t)^n$.*
3. *If $X \sim Geom(p)$, then $M_X(t) = \frac{pe^t}{1-(1-p)e^t}$ for $t < -\log(1-p)$.*
4. *If $X \sim Pois(\lambda)$, then $M_X(t) = e^{-\lambda(1-e^t)}$.*
5. *If $X \sim Exp(\lambda)$, then $M_X(t) = \frac{\lambda}{\lambda - t}$, for $t < \lambda$, and $+\infty$ otherwise.*
6. *If $X \sim Gamma(\alpha, \lambda)$, then $M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha$, for $t < \lambda$, and $+\infty$ otherwise*
7. *If $X \sim N(\mu, \sigma^2)$, then $M_X(t) = e^{t\mu + \frac{\sigma^2}{2}t^2}$.*

Proof. Some computations covered in class. □

Example 1 (moments of standard normal distribution)
Suppose that $Z \sim N(0, 1)$. Let's compute the moments $\mathbb{E}[Z^n],\ n \in \mathbb{N}$.

We have,

$$\sum_{n=0}^{+\infty} \frac{t^{2n}}{2^n n!} = e^{t^2/2} = M_Z(t) = \sum_{k=0}^{+\infty} \frac{t^k}{k!}\mathbb{E}[X^k].$$

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We compare the coefficients of t^k of the two power series and we find

$$\mathbb{E}[Z^n] = \begin{cases} 0, & \text{if } n \text{ is odd,} \\ \frac{n!}{2^{n/2}(n/2)!}, & \text{if } n \text{ is even.} \end{cases}$$

If two random variables X, Y have finite and equal moment generating functions, then they also have the same distribution. Therefore, by knowing the moment generating function of a random variable we, more or less, know its distribution. We make this statement more precise in the following theorem.

Theorem 13.4. *Let X, Y be two random variables. Suppose that there exists $\epsilon > 0$ such that*

1. *M_X, M_Y are finite in $(-\epsilon, \epsilon)$,*
2. *$M_X(t) = M_Y(t)$ for all $t \in (-\epsilon, \epsilon)$,*

then X, Y have the same distribution; that is $\mathbb{P}^X = \mathbb{P}^Y$.

Proof. We will prove this theorem in future lectures using *characteristic functions.* □

13.2 Sums of independent random variables

Example 2 (sum of independent and identically distributed random variables)
Suppose that $X_1, \ldots, X_n$ are independent and identically distributed random variables. Set $X = X_1 + \ldots + X_n$. We notice that

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\prod_{i=1}^n e^{tX_i}\right] = \prod_{i=1}^n \mathbb{E}[e^{tX_i}] = \prod_{i=1}^n M_{X_1}(t) = (M_{X_1}(t))^n.$$

If $X_1 \sim \text{Ber}(p)$ or $X_1 \sim \text{Exp}(\lambda)$, then the moment generating function of X is the one from Proposition 10.3 parts 2 and 6, respectively. Since these moment generating functions are finite, Theorem 10.4 yields

$$X_1 \sim \text{Ber}(p) \implies X = X_1 + \ldots X_n \sim \text{Bin}(n, p),$$
$$X_1 \sim \text{Exp}(\lambda) \implies X = X_1 + \ldots X_n \sim \text{Gamma}(n, \lambda).$$

Now assume that $X_1 \sim N(\mu, \sigma^2)$, then $M_X(t) = e^{tn\mu + \frac{n\sigma^2}{2}t^2}$, which is the moment generator function of a $N(n\mu, n\sigma^2)$ distribution. This means that

$$X_1 \sim N(\mu, \sigma^2) \implies X = X_1 + \ldots + X_n \sim N(n\mu, n\sigma^2)$$

Example 3 (sum of independent random variables, examples)
Suppose that $X_1, \ldots, X_n$ are independent random variables. We set $X = X_1 + \ldots + X_n$. Then, as in the previous example,

$$M_X(t) = M_{X_1}(t) \cdot \ldots \cdot M_{X_n}(t).$$

If $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1, \ldots, n$, then

$$M_X(t) = e^{t\sum_{i=1}^n \mu_i + \frac{t^2}{2}\sum_{i=1}^n \sigma_i^2}.$$

this implies that $X_1 + \ldots + X_n$ has a normal distribution with mean $\sum_{i=1}^n \mu_i$ and variance $\sum_{i=1}^n \sigma_i^2$.

In the above examples, we see that using the moment generating function, we may be able to find the distribution of a sum independent random variables. This is not always possible; for example how can we find the distribution of the sum of two independent random variables $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$? We will now see a general way to approach such a question.

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Definition 13.5 (Convolution). 1. Let $f, g : \mathbb{R} \to \mathbb{R}$ be integrable functions. The *convolution* of f and g is the function $f * g : \mathbb{R} \to \mathbb{R}$ defined by

$$(f * g)(x) := \int_{\mathbb{R}} f(x-y)\, g(y)\, dy, \qquad x \in \mathbb{R}.$$

2. Let μ and ν be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The *convolution* of μ and ν is the probability measure $\mu * \nu$ defined by

$$(\mu * \nu)(A) := \iint \mathbf{1}_A(x+y) d\mu(x) d\nu(y) = \int_{\mathbb{R}} \mu(A-y)\, d\nu(y) =: \mathbb{E}^{\nu}[\mu(A-y)], \qquad A \in \mathcal{B}(\mathbb{R}),$$

where $A - y := \{x - y : x \in A\}$ and $\mathbb{E}^{\nu}$ is the expected value on the probability space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \nu)$.

Suggested problem. Check that $f * g = g * f$ and $\mu * \nu = \nu * \mu$.

Theorem 13.6. *Suppose that X, Y are independent random variables with distributions $\mathbb{P}^X, \mathbb{P}^Y$. Recall that $\mathbb{P}^X, \mathbb{P}^Y$ are probability measures over $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then, $\mathbb{P}^{X+Y} = \mathbb{P}^X * \mathbb{P}^Y$.*

Proof. For any $A \in \mathcal{B}(\mathbb{R})$ [4], we have, by the law of total probability and the independence,

$$\mathbb{P}^{X+Y}(A) = \mathbb{P}(X+Y \in A) = \int \mathbb{P}(X+y \in A \mid Y = y) d\mathbb{P}^Y(y) = \int \mathbb{P}(X \in A-y) d\mathbb{P}^Y(y) = \mathbb{P}^X * \mathbb{P}^Y(A).$$

□

Corollary 13.7. *Suppose that X, Y are independent and that the distribution of X has density f_X. Then, the distribution of $X+Y$ has density*

$$f_{X+Y}(z) = \int f_X(z-y) d\mathbb{P}^Y(y).$$

Furthermore, if the distribution of Y has density f_Y, then the distribution of $X+Y$ has density $f_X * f_Y$.

Proof. By the previous Theorem, the distribution of $X+Y$ is $\mathbb{P}^X * \mathbb{P}^Y$. Since $\mathbb{P}^X$ has density f_X, we have, for any $A \in \mathcal{B}(\mathbb{R})$,

$$\begin{aligned}\mathbb{P}^X * \mathbb{P}^Y(A) &= \int \mathbb{P}(X \in A-y) d\mathbb{P}^Y(y) = \int \int_{A-y} f_X(x) dx d\mathbb{P}^Y(y) = \int \int_A f_X(x-y) dx d\mathbb{P}^Y(y) \\ &= \int_A \int f_X(x-y) d\mathbb{P}^Y(y) dx.\end{aligned}$$

This implies that the density of $\mathbb{P}^X * \mathbb{P}^Y$ is $\int f_X(x-y) d\mathbb{P}^Y(y)$. If $\mathbb{P}^Y$ has density f_Y, then this is equal to $\int f_X(x-y) f_Y(y) dy = f_X * f_Y(x)$. The proof is complete. □

Example 4
Assume that $X_1 \sim \mathrm{Exp}(\lambda_1)$ and $X_2 \sim \mathrm{Exp}(\lambda_2)$ with $\lambda_! \neq \lambda_2$ and that X_1, X_2 are independent.

Since X_1, X_2 are absolutely continuous, $X_1 + X_2$ is absolutely continuous with density

$$\begin{aligned}f_{X_1+X_2}(z) &= \int_{-\infty}^{+\infty} f_{X_1}(z-y) f_{X_2}(y) dy = \int_0^z \lambda_1 e^{-\lambda_1(z-y)} \lambda_2 e^{-\lambda_2 y} dy = \lambda_1 \lambda_2 e^{-\lambda_1 z} \int_0^z e^{y(\lambda_1 - \lambda_2)} dy \\ &= \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2} (e^{-\lambda_2 z} - e^{-\lambda_1 z}).\end{aligned}$$

[4] We can also just choose $A = (-\infty, z]$, for $z \in \mathbb{R}$.

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